

SERAFIMOVICH, Russian Federation

WMO: 343570

Lat: 49.567N

Lon: 42.733E

Elev: 423

StdP: 14.47

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.7	-2.2	-13.4	2.7	-6.7	-8.3	3.6	-1.1	27.4	29.5	25.0	25.6	4.9	290	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	95.0	68.5	91.3	67.1	87.8	66.1	70.9	87.7	69.3	85.5	68.0	82.8	7.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.8	96.8	75.5	64.3	91.5	74.8	62.6	86.4	74.0	34.9	88.0	33.7	85.9	32.5	82.5	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.9	20.2	17.6	-11.2	98.2	8.4	4.0	-17.2	101.1	-22.1	103.4	-26.8	105.7	-32.9	108.6		
			-12.3	73.1	8.6	2.5	-18.5	74.9	-23.5	76.3	-28.3	77.7	-34.5	79.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	20.7	22.2	32.9	50.0	62.8	70.1	74.3	72.7	60.6	47.6	34.5	25.2	
	DBStd	21.37	11.52	11.73	8.85	8.53	8.54	7.25	6.57	7.56	7.56	8.77	9.57	10.62	
	HDD50	3723	907	778	530	104	8	0	0	0	9	150	469	768	
	HDD65	7080	1372	1198	994	453	147	29	6	20	173	541	914	1233	
	CDD50	2975	0	0	1	103	405	603	753	703	326	77	4	0	
	CDD65	856	0	0	0	2	78	182	294	258	40	2	0	0	
	CDH74	8472	0	0	0	24	720	1727	3019	2666	304	12	0	0	
	CDH80	3490	0	0	0	1	225	694	1339	1175	54	2	0	0	
Wind	WSAvg	6.7	7.5	8.5	8.6	7.7	6.4	5.5	4.9	5.2	5.7	6.2	7.2	7.6	
Precipitation	PrecAvg	17.1	1.3	1.1	1.0	1.2	1.7	2.0	1.8	1.6	1.5	1.1	1.6	1.7	
	PrecMax	24.5	3.8	3.0	4.6	6.0	5.6	5.0	4.3	5.0	4.6	3.0	4.4	4.7	
	PrecMin	8.3	0.2	0.0	0.0	0.0	0.2	0.2	0.1	0.0	0.1	0.0	0.2	0.5	
	PrecStd	4.3	0.8	0.7	0.8	1.0	1.3	1.3	1.3	1.2	1.1	0.8	1.1	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	45.5	58.8	78.0	90.8	96.7	99.4	99.9	85.3	74.6	59.4	46.6	
		MCWB	41.7	42.4	47.0	57.7	63.4	67.4	68.9	70.0	63.2	60.8	52.7	45.0	
	2%	DB	39.6	40.3	52.9	72.4	86.0	91.7	94.9	94.8	81.4	68.6	54.0	43.3	
		MCWB	38.2	37.8	44.2	56.1	62.9	66.9	68.8	68.3	62.1	57.2	49.5	41.7	
	5%	DB	36.5	37.4	48.6	67.6	81.6	87.4	91.3	90.6	77.7	64.2	50.3	40.3	
		MCWB	34.9	35.5	42.2	53.7	61.5	65.8	67.8	66.8	60.9	55.5	47.2	38.5	
	10%	DB	34.5	35.5	44.7	63.9	77.5	83.7	87.5	86.5	73.6	60.0	46.9	37.5	
		MCWB	33.2	33.9	40.0	51.4	60.5	64.7	66.9	65.8	59.4	53.0	44.2	36.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	42.3	49.2	60.2	67.3	71.7	72.8	73.2	67.6	63.1	53.4	45.2
			MCDB	43.6	44.9	55.7	73.1	83.7	88.7	88.7	93.3	77.2	71.6	57.0	46.3
2%		WB	38.4	38.2	45.6	57.2	65.1	69.6	71.0	70.4	64.8	59.3	50.4	41.7	
		MCDB	39.7	39.8	51.4	69.2	80.5	84.4	87.4	88.8	75.3	66.1	53.5	43.0	
5%		WB	35.2	35.9	42.8	55.1	63.5	67.9	69.7	68.5	62.7	56.2	47.5	38.6	
		MCDB	36.3	37.1	47.8	65.2	77.6	82.8	86.1	85.7	73.1	62.0	50.1	40.0	
10%		WB	33.3	34.1	40.2	52.9	61.6	66.3	68.4	66.8	60.8	53.5	44.5	36.1	
		MCDB	34.3	35.3	44.2	61.9	74.9	80.2	82.8	82.6	71.5	59.4	46.6	37.3	
Mean Daily Temperature Range		5% DB	MDBR	8.9	10.0	11.8	16.4	18.0	17.7	18.6	19.9	17.4	13.0	9.1	8.2
			MCDBR	6.7	8.8	17.5	21.3	23.1	23.0	23.8	24.6	22.7	19.0	11.9	8.1
	MCWBR		6.3	7.4	11.0	10.1	9.3	8.5	8.5	9.3	10.3	10.6	9.1	7.3	
	5% WB	MCDBR	6.7	7.8	15.5	18.8	20.2	20.1	20.0	21.8	18.4	16.5	10.8	7.5	
		MCWBR	6.4	7.1	10.5	9.7	9.2	8.3	8.2	9.5	9.7	11.0	8.9	7.2	
Clear-Sky Solar Irradiance	taub		0.319	0.352	0.379	0.440	0.407	0.399	0.422	0.428	0.394	0.363	0.339	0.320	
	taud		2.122	2.029	2.115	2.085	2.276	2.336	2.278	2.238	2.304	2.340	2.289	2.183	
	Ebn at Noon		218	239	258	254	270	273	264	256	252	239	214	200	
	Edh at Noon		26	37	41	47	40	38	40	40	34	27	23	22	
All-Sky Solar Radiation	RadAvg		293	613	991	1394	1852	2012	1947	1756	1235	680	303	185	
	RadStd		67	126	141	131	176	195	125	119	115	95	59	38	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+1.01	N/A	N/A	N/A	N/A	N/A	N/A	-254	+171	N/A

Nomenclature: See separate page